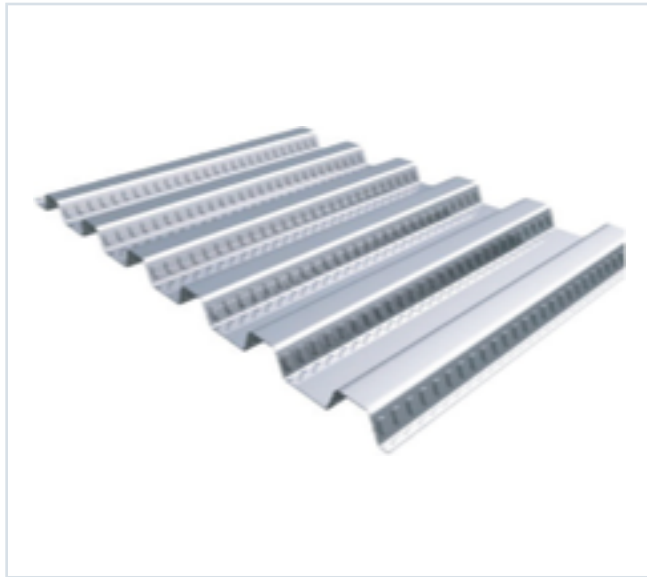




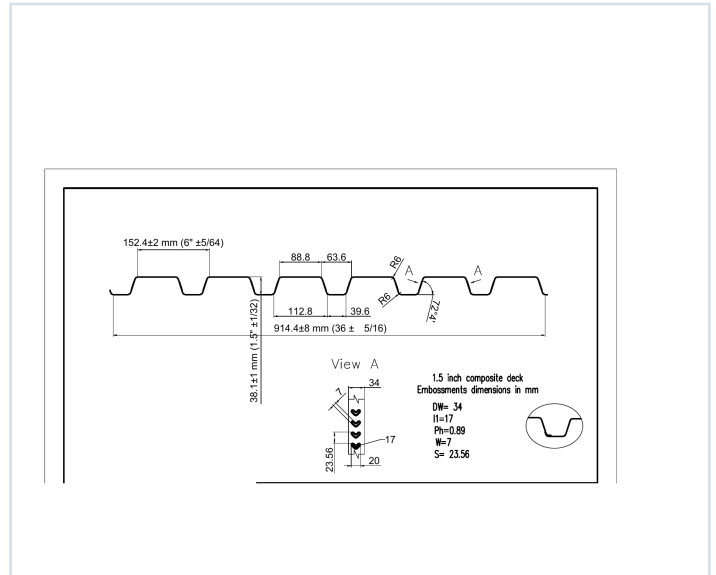
DECKING PRODUCT SUBMITTAL

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 80	18 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in³/ft)	S- (in³/ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in⁴/ft)	I- (in⁴/ft)
0.0474	2.6	60	0.291	0.306	10455	11006	0.252	0.288

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 50, 51, 52, 53
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.



Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

15CD-FY80-18-G90 | Page 2

PUBLISHED PERFORMANCE DATA

15CD-FY80-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 52

22 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	375	374	209	232	202
4	37	400	400	400	397	340	293	238
4.5	43	400	400	400	400	400	397	311
5	49	400	400	400	400	400	400	369
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	177	166	138	123	110	98	89	80
4	224	187	175	166	159	125	113	102
4.5	273	241	213	190	170	153	138	125
5	323	289	255	228	202	182	164	148
5.5	375	331	294	262	235	211	191	175
6	400	378	336	300	269	242	218	197

Note
ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	400	382	327	283	246
4	37	400	400	400	400	400	368	312
4.5	43	400	400	400	400	400	400	381
5	49	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	236	181	170	161	136	122	110	100
4	276	242	215	192	172	155	142	127
4.5	333	296	263	236	211	190	172	156
5	390	352	313	280	251	226	205	186
5.5	450	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 gage. However, the construction spans do get longer for 18 and 16 gage deck.
2. ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

STEEL CATALOG PAGE 53

22 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	359	374	260	225	196
4	28	400	400	400	386	329	285	248
4.5	33	400	400	400	400	400	348	304
5	37	400	400	400	400	400	400	362
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	172	162	135	121	108	97	88	80
4	208	183	172	163	158	124	112	102
4.5	267	236	210	188	168	152	137	125
5	318	285	250	224	201	181	164	149
5.5	369	327	291	260	234	211	191	175
6	400	374	333	298	268	242	218	199

Note
ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	400	367	314	272	238
4	28	400	400	400	400	398	346	302
4.5	33	400	400	400	400	400	400	370
5	37	400	400	400	400	400	400	400
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	279	181	161	147	132	120	108	99
4	306	235	209	187	169	152	136	125
4.5	325	288	257	230	207	187	169	154
5	368	344	306	275	247	223	202	184
5.5	400	400	357	320	288	260	236	215
6	400	400	400	366	330	298	271	246

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 gage. However, the construction spans do get longer for 18 and 16 gage deck.
2. ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.